

Learning Curve Extrapolation Across Scales

Study Project

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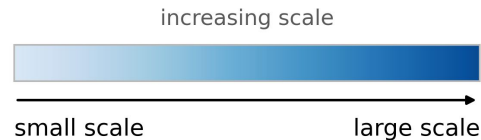
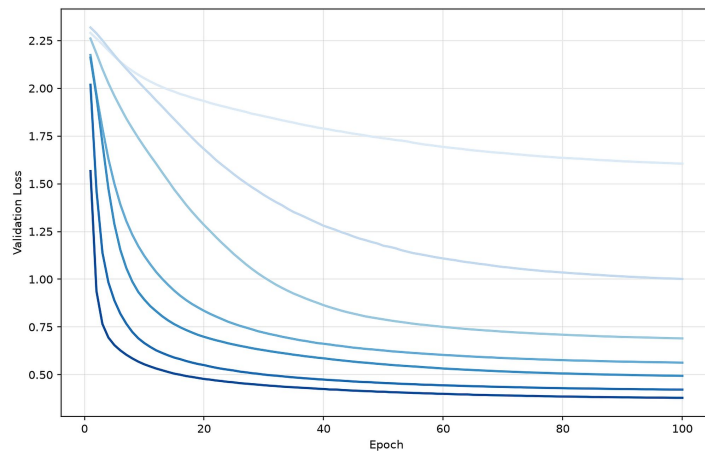
Motivation



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Why this matters

- Training runs at larger scale are costly - testing every hyperparameter combination directly at full scale isn't practical
- Scaling laws show model performance follows predictable trends with size
- This project asks whether we can use a small-scale training curve to extrapolate a larger-scale one



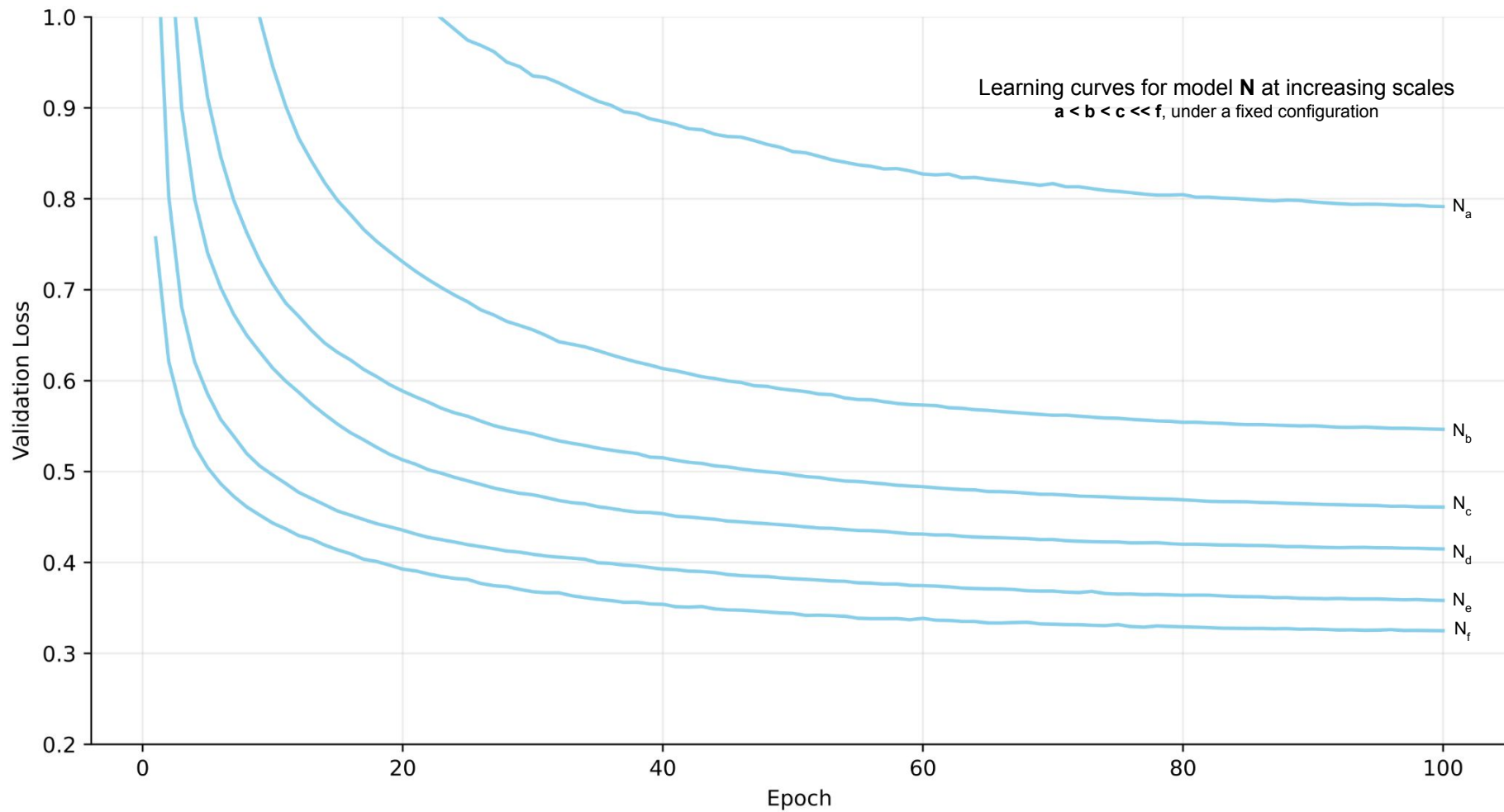
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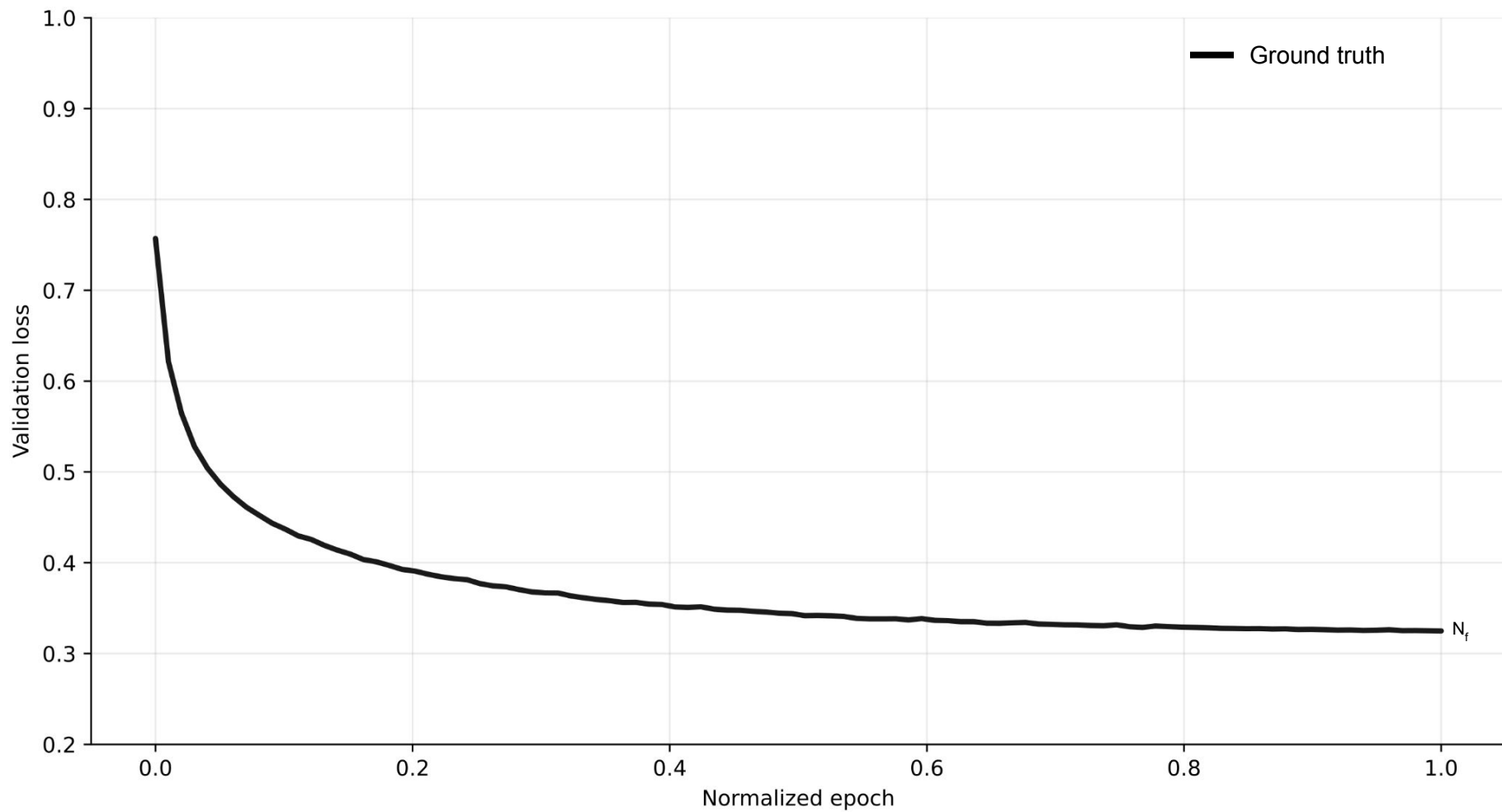
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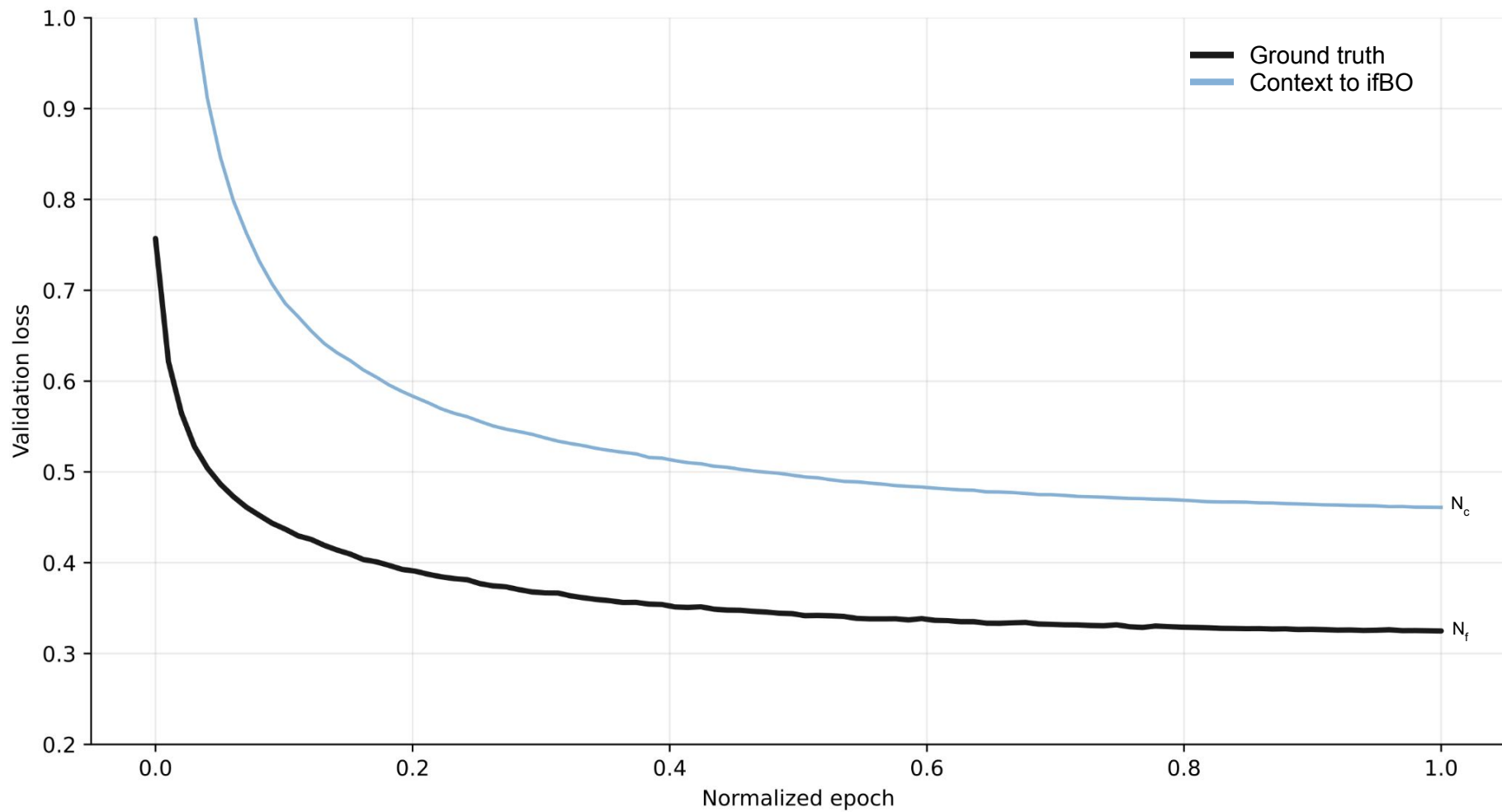
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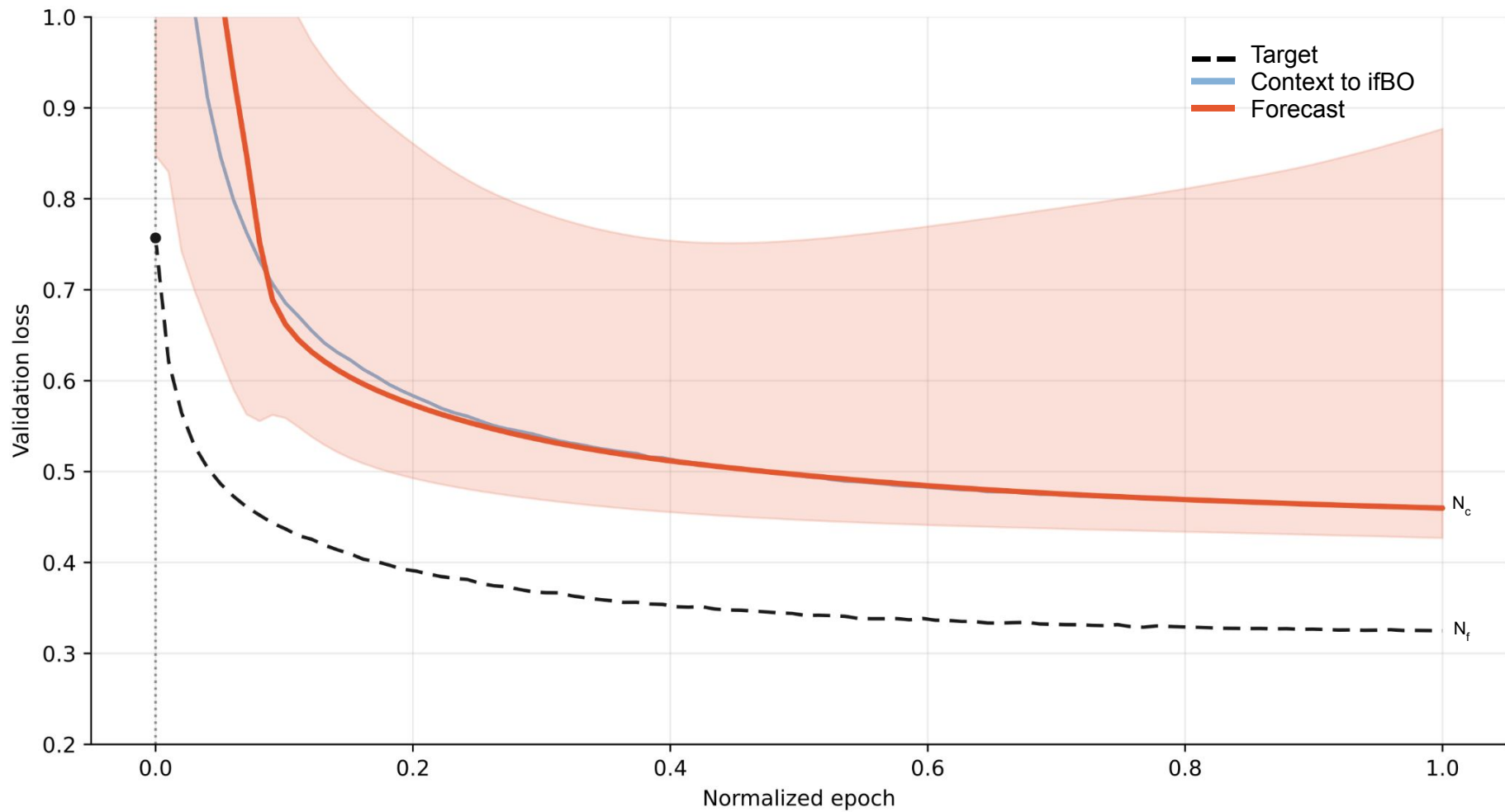
Why ifBO

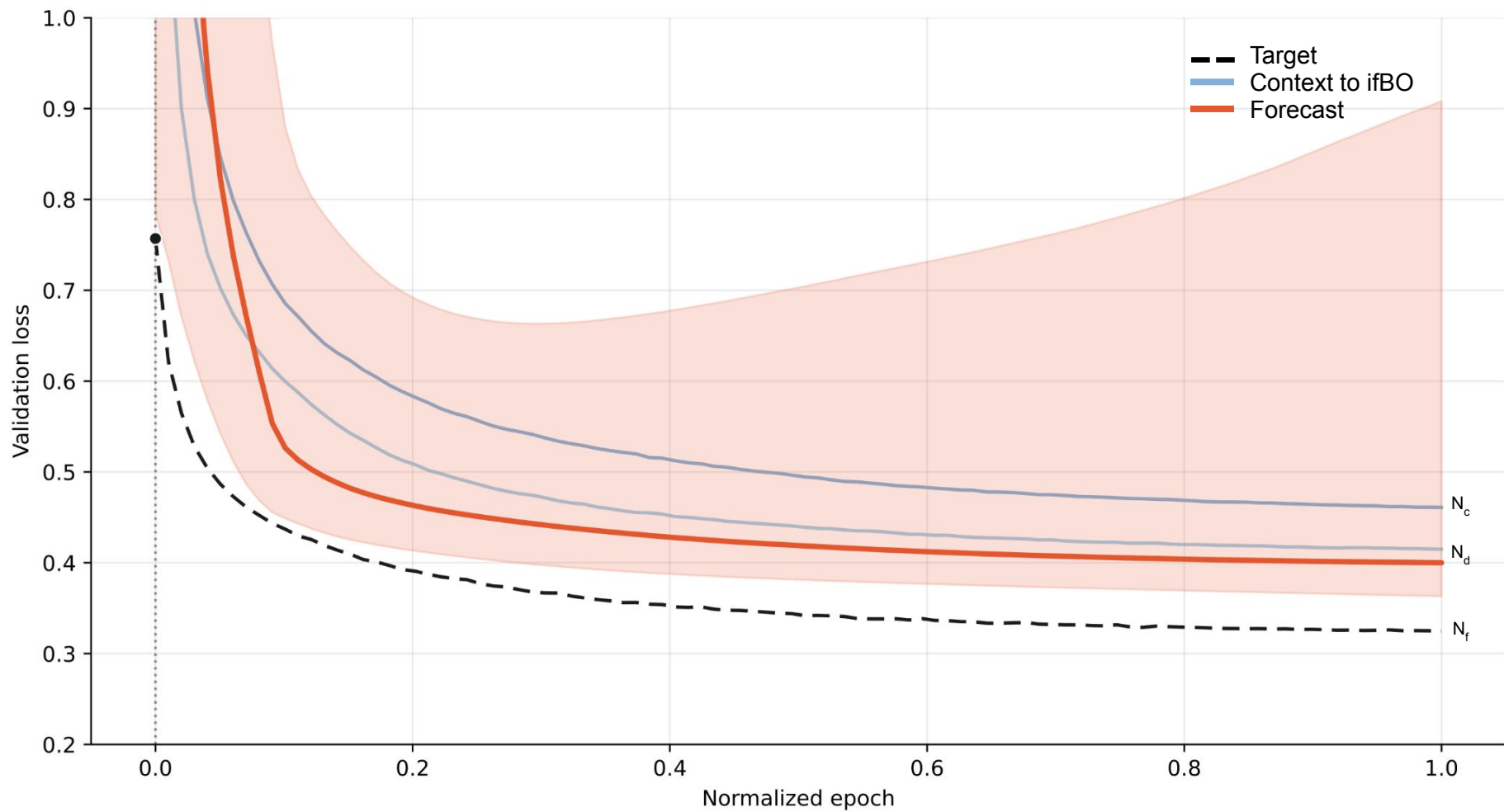
- Learns from context curves on the fly - no retraining or refitting needed for new datasets or configurations
- Outputs a full predictive distribution, not just a point estimate - useful for deciding when to stop a run early

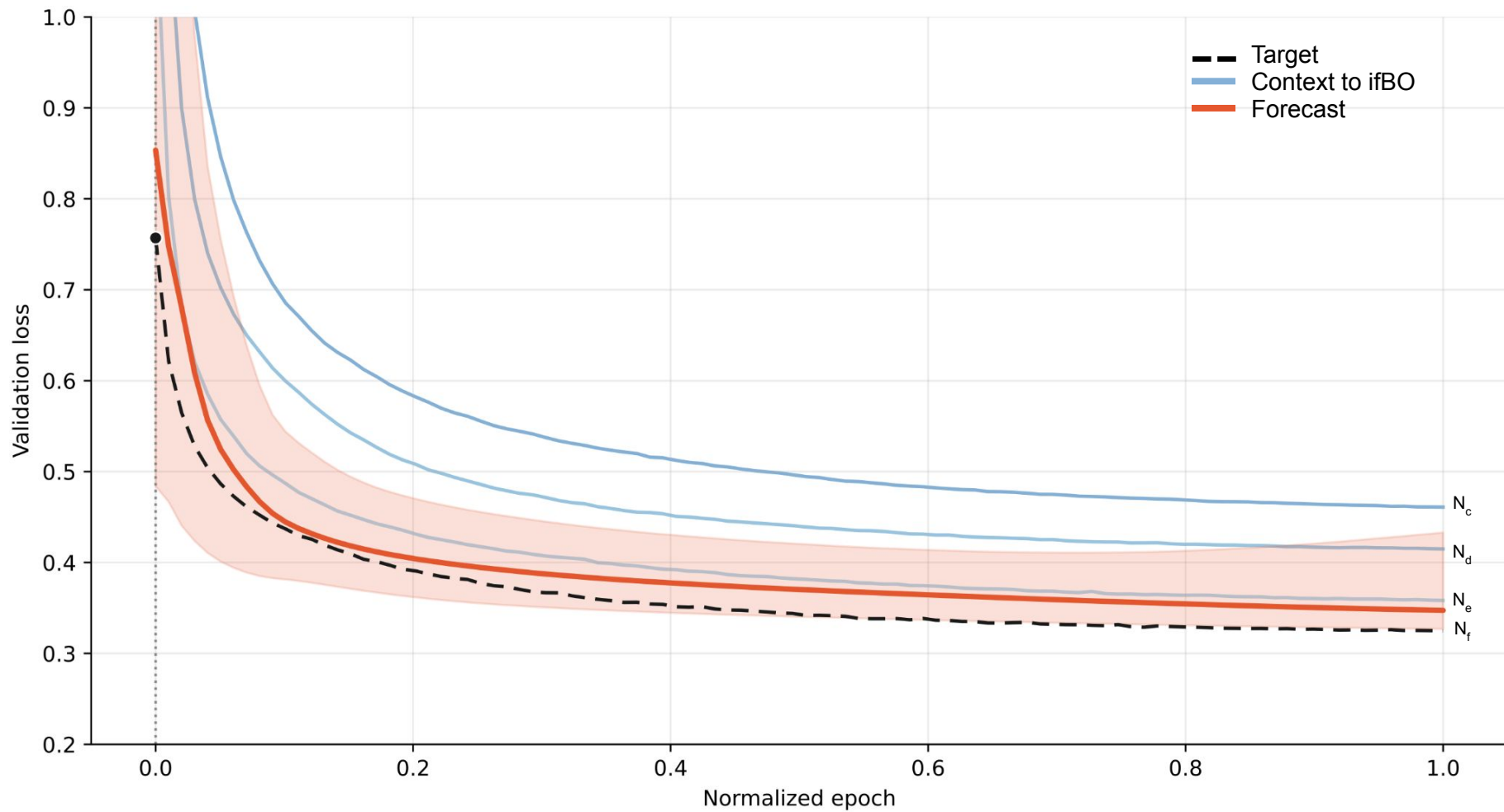


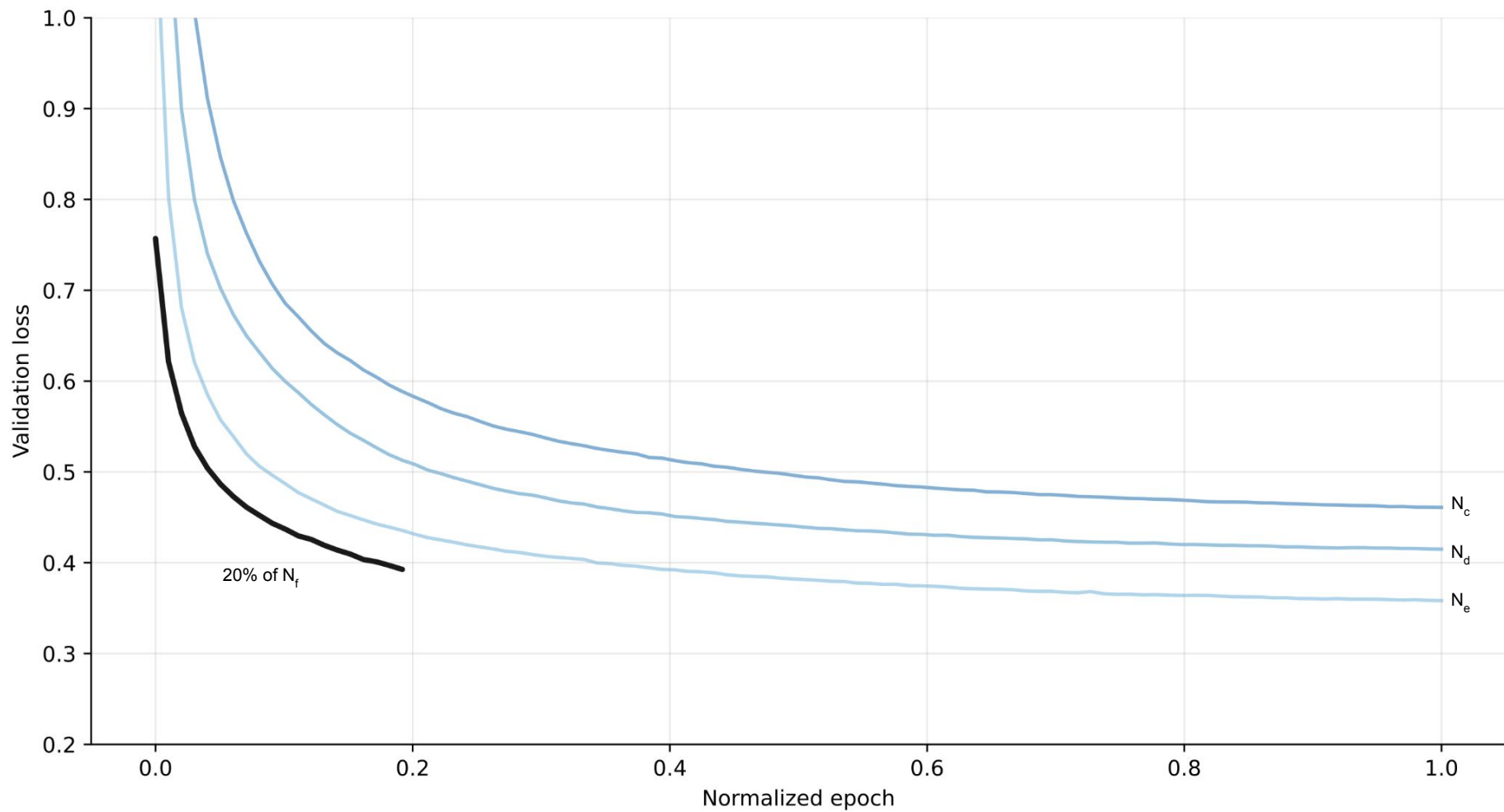


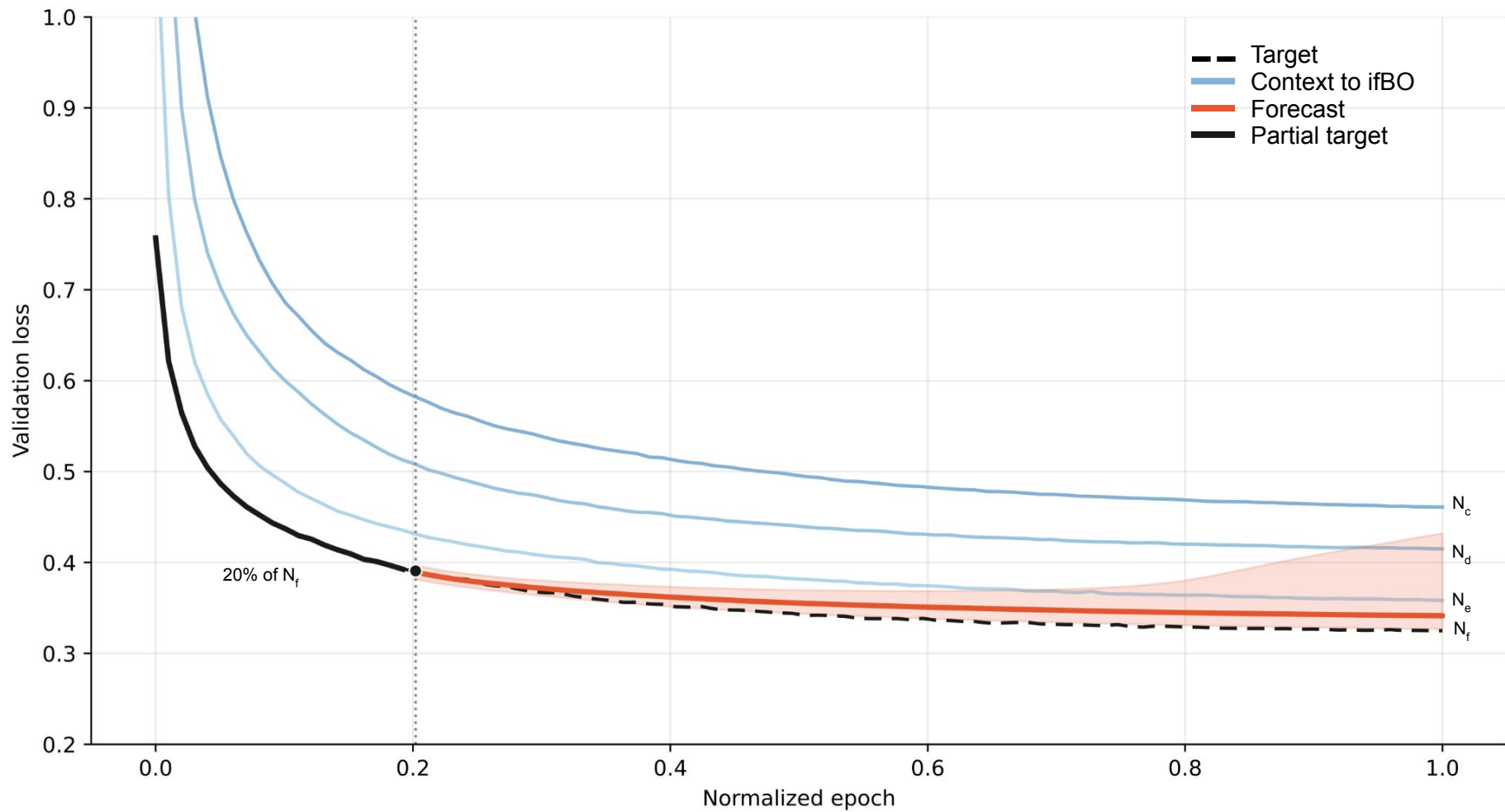


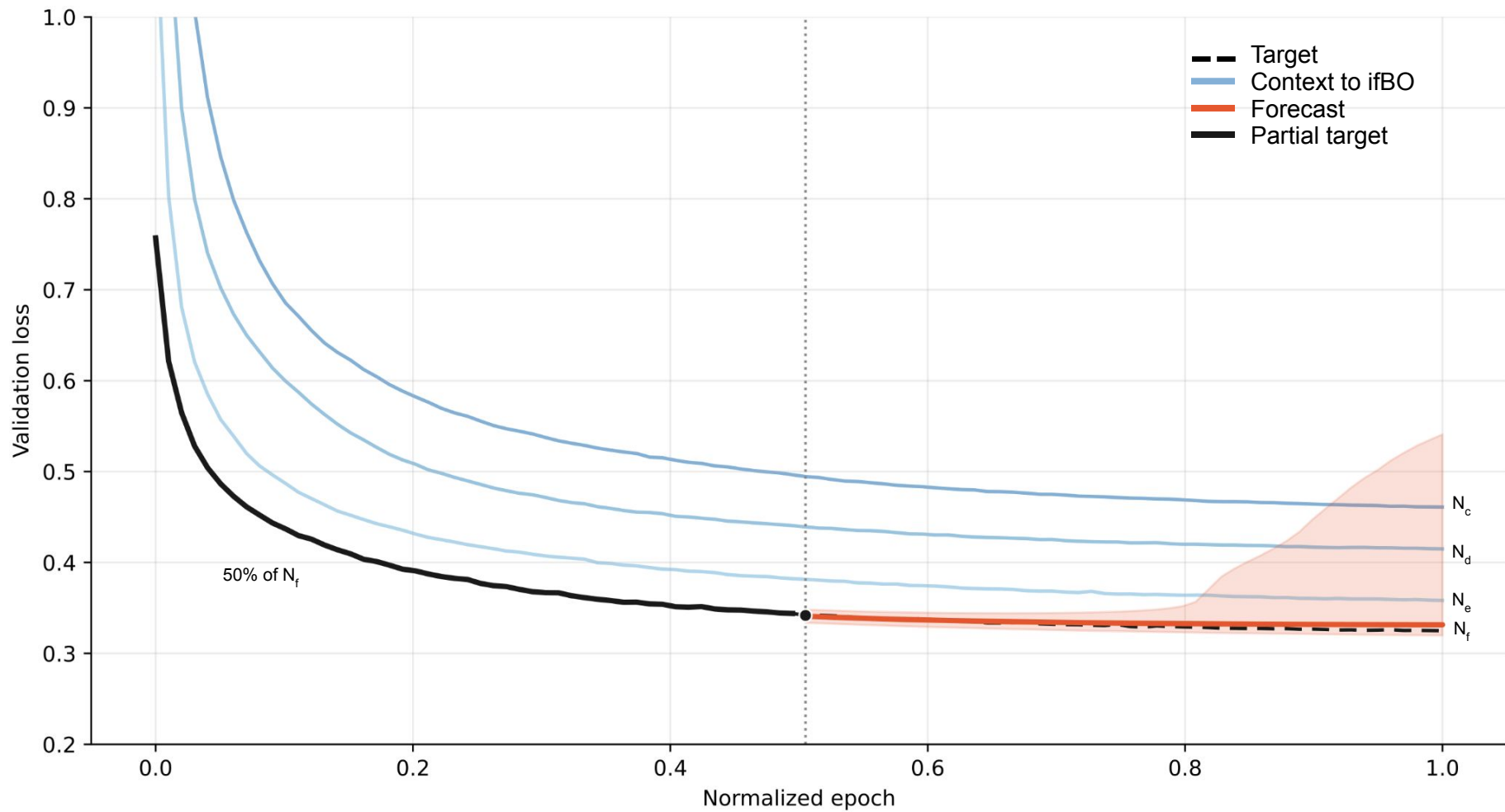


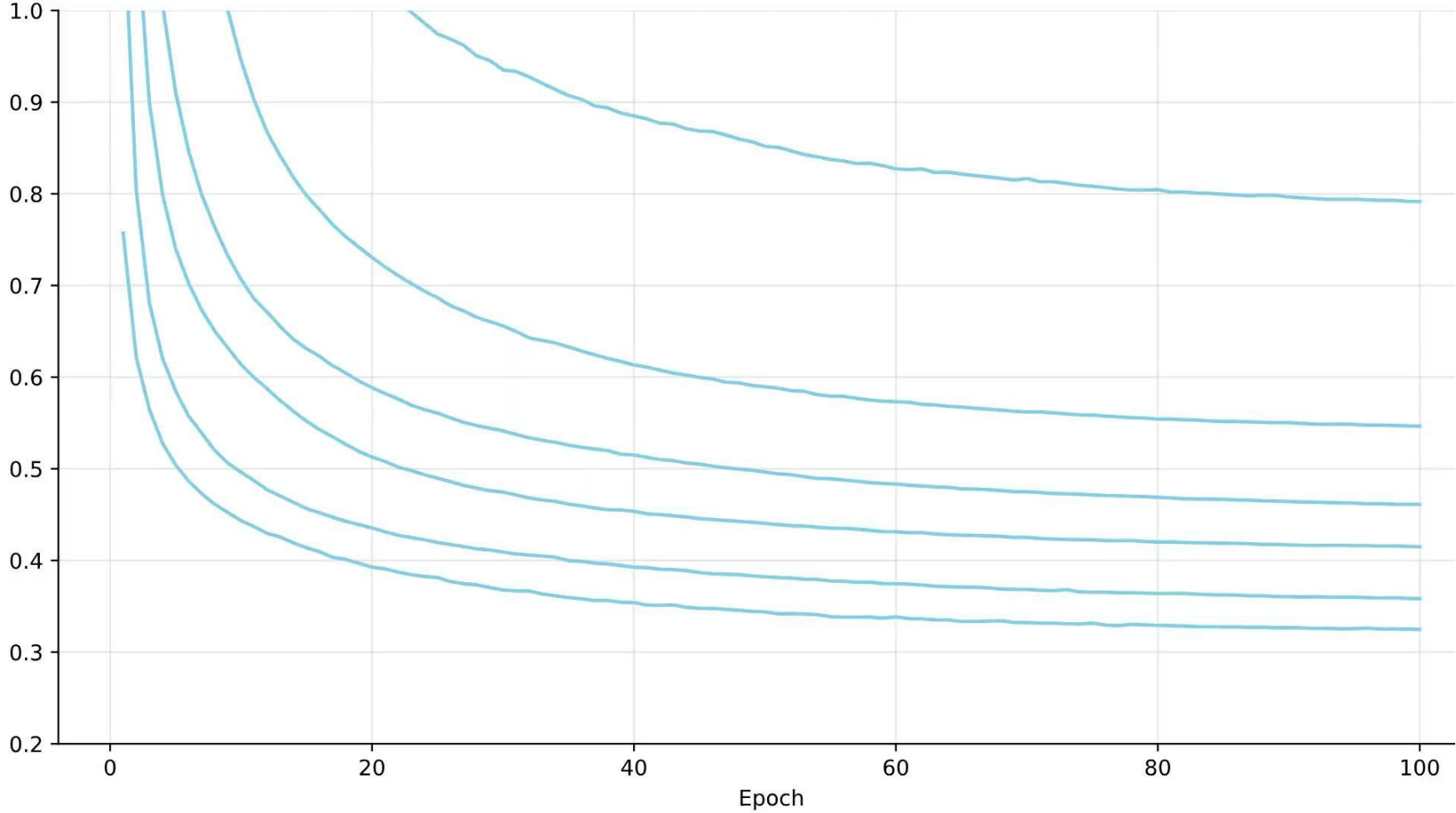












Setup

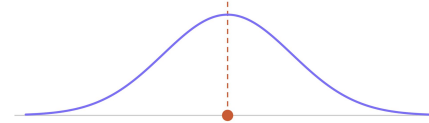


Setup

- Datasets: Toy, real world
- Metric
 - NLL
 - Relative error %
 - WIS - Weighted Interval score



NLL



Relative error %

$$\text{Rel err \%} = \frac{|\text{pred}_{\text{final}} - \text{truth}_{\text{final}}|}{\text{truth}_{\text{final}} + \varepsilon} \times 100$$

WIS



Method



Method - Round Robin

01

Order lower scales.

- Collect for every distinct **N** it's corresponding low scales
- Order by nearest-first

02

Rank within each scale.

- Normalize each scale's HP vectors
- Rank configurations by **Euclidean distance** to the query's HP vector

03

Round-robin.

- Take rank **1** from each scale
- Then rank **2** from each, and so on
- Stop once **k** curves are collected

Observations



Influence of adding smaller scale curves

- k=12 gives nearly 2x better relative error than baseline
- We don't need every curve from lower scales
- Randomly choosing curves hurt more than feeding all lower curves

POLICY_VARIANT	K	REL_ERR%	NLL	WIS
Round robin	16	7.31	-0.74	0.0594
Round robin	12	7.10	-0.80	0.0571
Round robin	8	7.24	-0.63	0.0591
Round robin	4	7.68	-0.69	0.0677
All lower	-	7.30	-0.44	0.0579
Random	12	7.96	-0.51	0.0685
Random	4	8.74	-0.49	0.0857
Baseline	-	16.21	0.29	0.1986

All values (lower=better)

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- Adding every curve from lower scales does improve prediction
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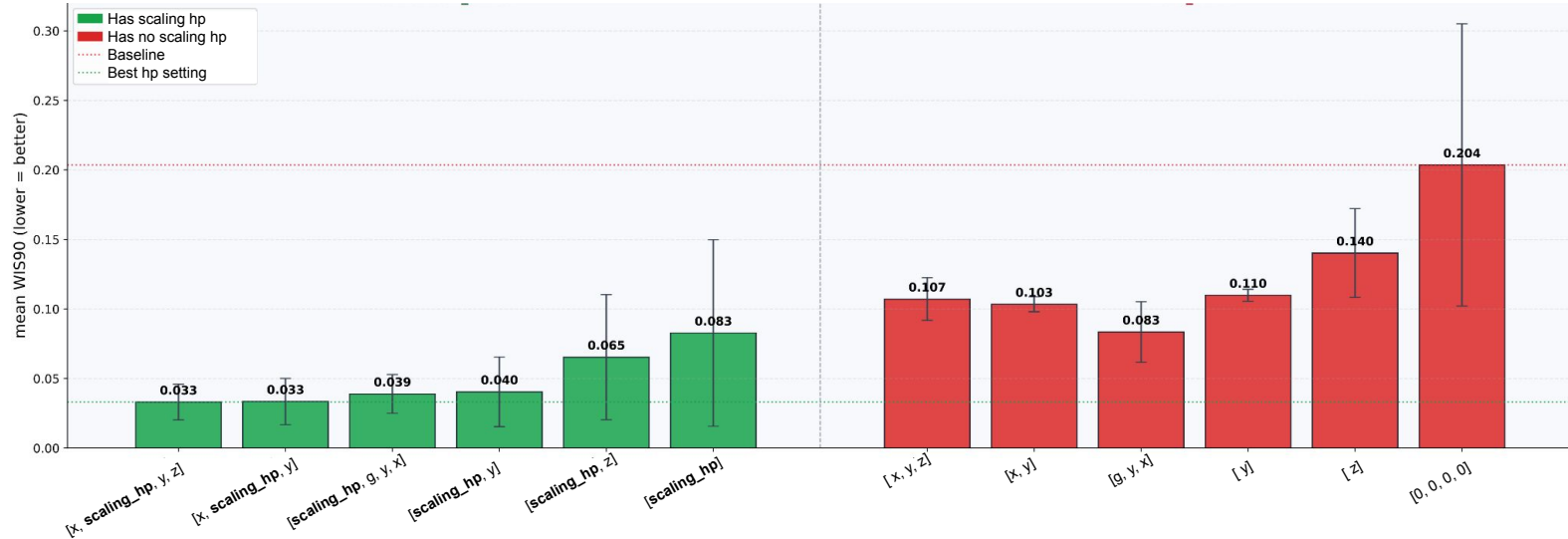
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Importance of scaling as a hyperparameter

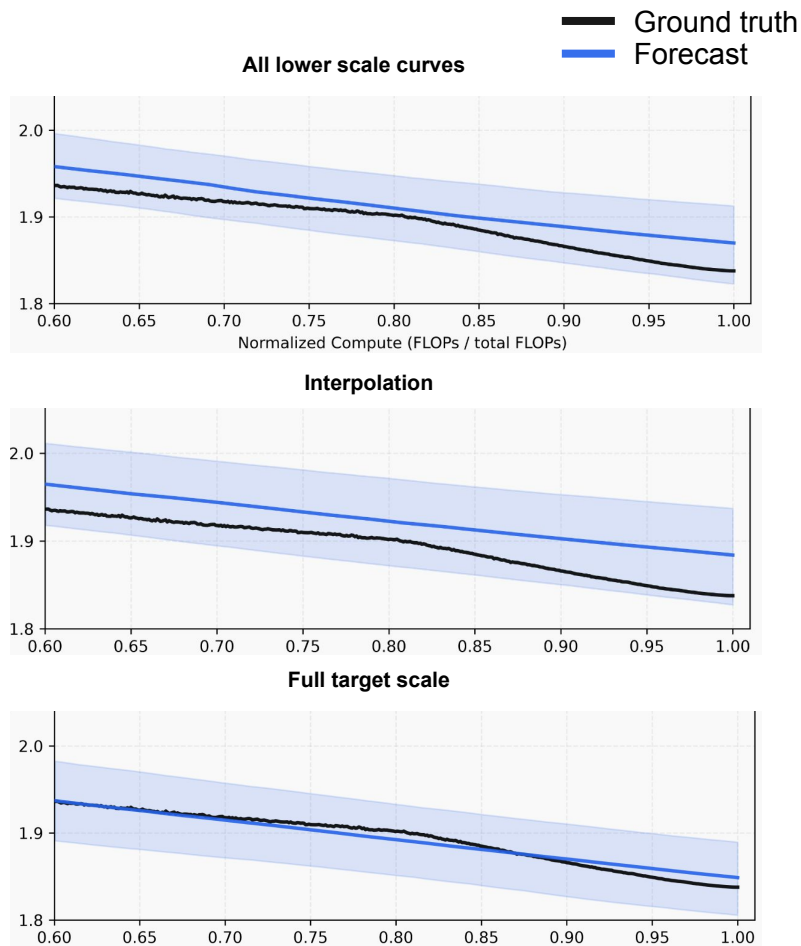


- Adding scaling hyperparameter consistently improves prediction accuracy (green bars beat red bars)
- **6x** better than the baseline
- **Scaling matters** more than other hyperparameters included alongside it

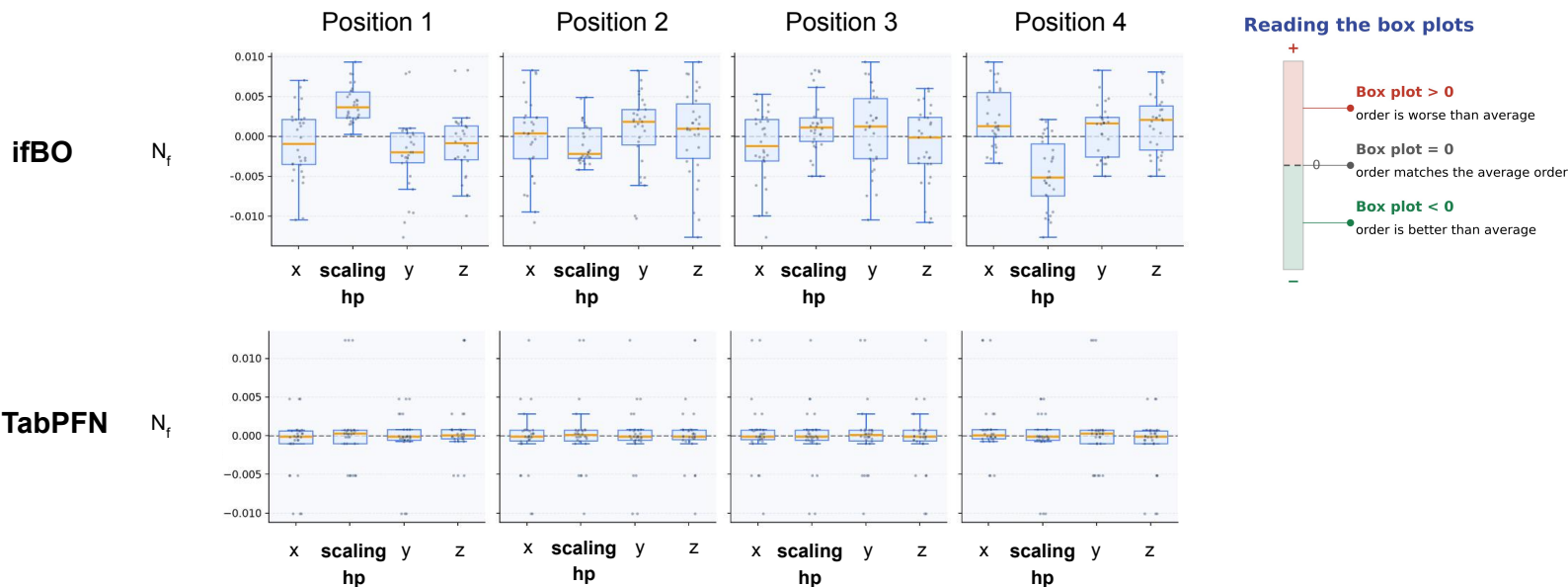
Predicting the shape of the curve

- All methods fit the observed region well
- But predicts a slowly declining line instead of following the true downward bend after **0.8**
- Even interpolation and full target scale, can't fully capture that late drop

NOTE: "More observed target data leads to better final point forecasting."

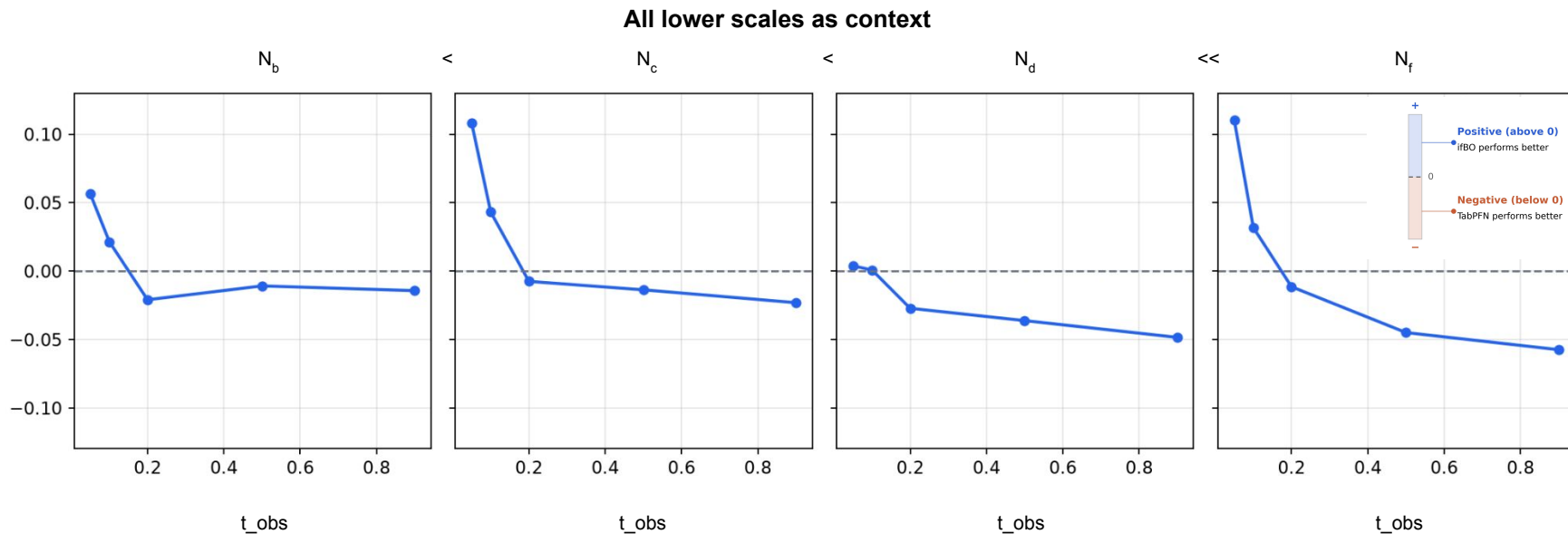


Invariance to HP ordering



- ifBO shows more spread and shifting medians, especially for scaling hyperparameter, indicating **order sensitivity**
- TabPFN stays near zero across all positions - largely order-invariant

Is ifBO any better than TabPFN?



- ifBO wins early, but TabPFN takes over as more of the target is observed
- As the distance between the target and context grows, still ifBO performs very well at early stages

Conclusion

- Extrapolating small-scale curves to larger scales is feasible
- Which hyperparameters you include matters more than how many curves you use
- Predictions fit the observed region well but miss late-stage bends, even with full context - extrapolation has a ceiling
- ifBO is also sensitive to hyperparameter ordering, unlike TabPFN - a practical robustness gap
- ifBO beats Tab PFN early, but TabPFN catches up and overtakes as more of the target is observed
- Overall, ifBO offers real value for early-stage forecasting, but isn't a clear winner across the board - the right choice depends on how much of the curve you've already observed.

References

Scaling law: Kaplan et al. (2020). *Scaling Laws for Neural Language Models*.

ifBO: Rakotoarison et al. (2024). *In-Context Freeze-Thaw Bayesian Optimization for HPO*. ICML 2024.

WIS: Bracher et al. (2021). *Evaluating Epidemic Forecasts in an Interval Format*. PLOS Comp. Biology.

TabPFN: Hollmann et al. (2023). *TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second*. ICLR 2023.

Thank you



Appendix



Weighted Interval Score

Weighted Interval Score (WIS)

Interval Score (IS) for interval k with level α_k :

$$IS_{\alpha_k}(y) = (u_k - l_k) + \frac{2}{\alpha_k} \max(l_k - y, 0) + \frac{2}{\alpha_k} \max(y - u_k, 0)$$

Weighted Interval Score (general form, K intervals)

$$WIS(y) = \frac{1}{K + \frac{1}{2}} \left[w_0 |y - \hat{m}| + \sum_{k=1}^K w_k IS_{\alpha_k}(y) \right]$$

$w_0 = \frac{1}{2}, \quad w_k = \frac{\alpha_k}{2}$

where:

- y = True (ground-truth) value
- $\hat{m}$ = Predicted median
- K = Number of prediction intervals used
- α_k = Significance level of interval k
- l_k, u_k = Lower / upper bound of interval k
- w_0, w_k = Weights: $w_0 = \frac{1}{2}, w_k = \frac{\alpha_k}{2}$

Method

1 · THE QUERY

TARGET_N = 610.5M base_N 14.6M n_embd 2048 shrink 0.2 tkpm 10

k = 4 **8** 12 16 24 32 2 round-robin pass(es) needed

■ target_N = 287.2M 35 candidates, ranked by HP distance

D=0.36 sh 0.2 tk 10 e 1280	D=0.41 sh 0.4 tk 10 e 1280	D=0.50 sh 0.2 tk 10 e 1280	D=0.54 sh 0.4 tk 10 e 1280	D=0.55 sh 0.2 tk 20 e 1280	D=0.55 sh 0.6 tk 10 e 1280	+29 more
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■ target_N = 134.6M 27 candidates, ranked by HP distance

D=0.59 sh 0.2 tk 10 e 768	D=0.63 sh 0.4 tk 10 e 768	D=0.69 sh 0.2 tk 10 e 768	D=0.72 sh 0.4 tk 10 e 768	D=0.73 sh 0.2 tk 20 e 768	D=0.76 sh 0.4 tk 20 e 768	+21 more
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■ target_N = 77.1M 24 candidates, ranked by HP distance

D=0.71 sh 0.2 tk 10 e 512	D=0.74 sh 0.4 tk 10 e 512	D=0.79 sh 0.2 tk 10 e 512	D=0.82 sh 0.4 tk 10 e 512	D=0.83 sh 0.2 tk 20 e 512	D=0.85 sh 0.4 tk 20 e 512	+18 more
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■ target_N = 32.3M 15 candidates, ranked by HP distance

D=0.83 sh 0.2 tk 10 e 256	D=0.86 sh 0.4 tk 10 e 256	D=0.93 sh 0.2 tk 20 e 256	D=0.93 sh 0.6 tk 10 e 256	D=0.95 sh 0.4 tk 20 e 256	D=1.02 sh 0.6 tk 20 e 256	+9 more
------------------------------------	------------------------------------	------------------------------------	------------------------------------	------------------------------------	------------------------------------	---------

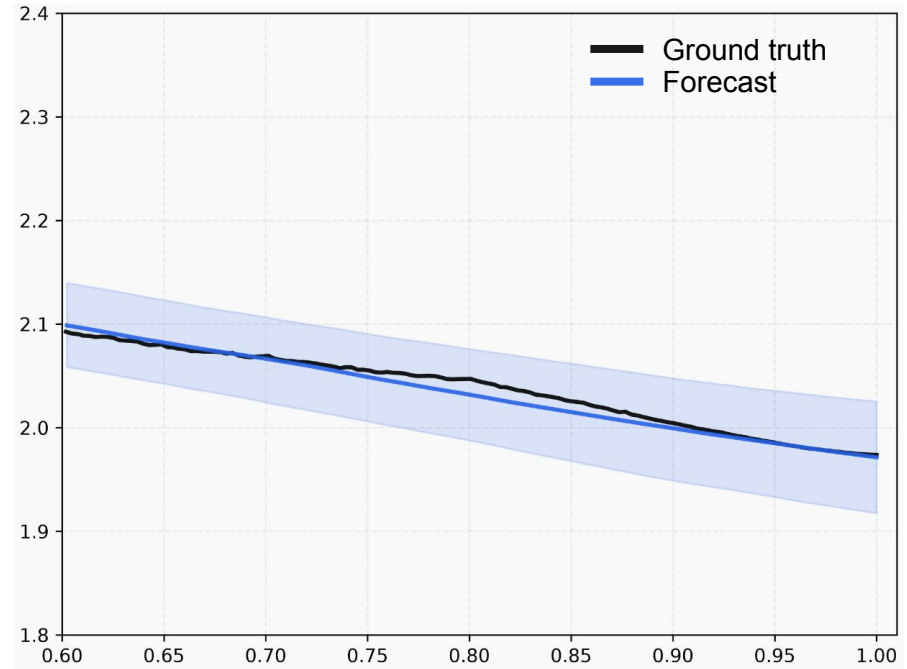
CONTEXT SET, PICK ORDER → 1 2 3 4 5 6 7 8

Toy data Observation

POLICY_VARIANT	K	REL_ERR %	NLL	WIS	WIS90
diverse_lower_k k=4	4	6.22	-0.87	0.0177	0.0160
diverse_lower_k k=8	8	5.60	-0.49	0.0162	0.0152
diverse_lower_k k=12	12	5.57	-0.90	0.0159	0.0147
diverse_lower_k k=16	16	5.66	-0.88	0.0166	0.0152
diverse_lower_k k=24	24	5.48	-0.80	0.0161	0.0148
diverse_lower_k k=32	32	5.66	-0.74	0.0168	0.0154
all_lower	-	5.95	-0.65	0.0178	0.0162
nearest_lower	-	6.81	-0.68	0.0206	0.0195
baseline	-	14.33	-0.99	0.0435	0.0377
interpolation	-	4.81	-2.08	0.0132	0.0118

Final Point Prediction Accuracy

- With more observed target data the final point prediction improves



INVARIANCE TO ORDERING

- For one full predicted curve, average over all predicted points:

$$\text{WIS}_{\text{curve}} = \frac{1}{T} \sum_{t=1}^T \text{WIS}_t$$

- For one ordering (r), scale (N), and observation level (o), average across trajectories:

$$S_m(N, o, r) = \frac{1}{|\mathcal{T}_{N,o}|} \sum_{\tau \in \mathcal{T}_{N,o}} \text{WIS}_{\text{curve}}(\tau, r)$$

Invariance to Ordering

-

$$\bar{S}_m(N, o) = \frac{1}{24} \sum_{r \in \mathcal{R}} S_m(N, o, r)$$

- where $(\mathcal{R})$ is the 24 HP permutations.
- Then the plotted centered value is:

-

$$\Delta S_m(N, o, r) = S_m(N, o, r) - \bar{S}_m(N, o)$$

-

$$S_m(N, o, r) = \bar{S}_m(N, o)$$

- $(\Delta S_m < 0)$: this order is better than the model's average order.
- $(\Delta S_m = 0)$: this order matches the model's average order.
- $(\Delta S_m > 0)$: this order is worse than the model's average order.

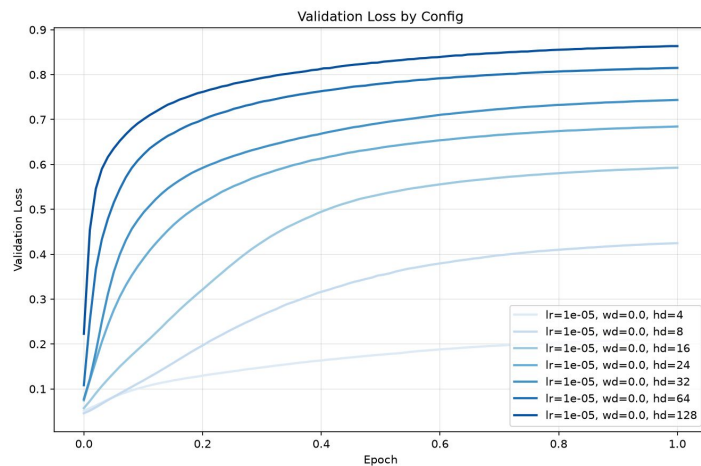
- So:

- **Per HP box:** $(5 \times 6 = 30)$ points
- **Per subplot:** $(4 \times 30 = 120)$ points
- **Per scale row across all 4 positions:** $(4 \times 120 = 480)$ plotted dots

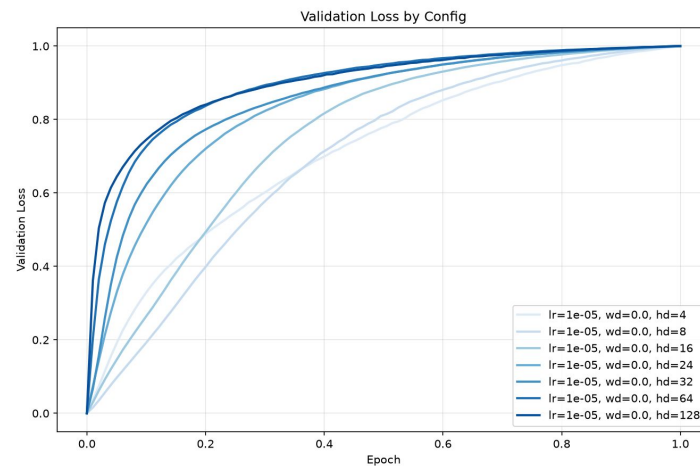
Normalization of y (Validation loss)

Types we considered

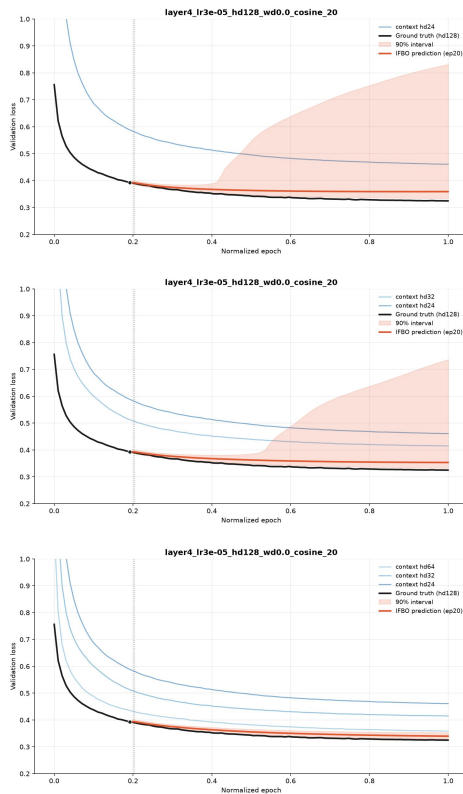
Global Normalization



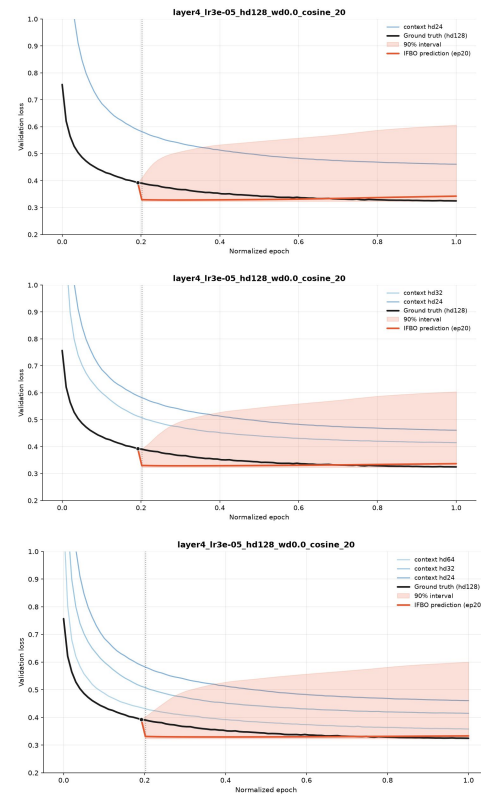
Per-curve Normalization



Global normalization



Per curve normalization



Added lower scales of the same configuration as context to predict a target scale (Method 1)

Subsampling

Types we considered

- **For y (validation loss)**
 - Linspace / Uniform
 - Geometric spacing
 - 3 zone sampling

We kept subsampling of 10 points from a curve for our experiments